Table 2. Predictive performance of the seven candidate models in internal and external evaluation.

A. Internal validation (133 patients; 55 events)

Model	AUROC (95% CI)	Average precision (95% CI)	Brier score (95% CI)	CITL (95% CI)	Calibration slope (95% CI)
Logistic regression	0.762 (0.672–0.842)	0.716 (0.589–0.823)	0.193 (0.160–0.231)	-0.063 (-0.493–0.355)	0.796 (0.480–1.268)
LASSO	0.766 (0.676–0.843)	0.721 (0.597–0.825)	0.192 (0.164–0.224)	-0.047 (-0.443–0.329)	1.054 (0.652–1.642)
Elastic net	0.763 (0.674–0.841)	0.716 (0.593–0.819)	0.193 (0.167–0.223)	-0.027 (-0.411–0.334)	1.226 (0.770–1.877)
Random forest	0.762 (0.672–0.842)	0.671 (0.540–0.802)	0.192 (0.159–0.230)	-0.092 (-0.520–0.313)	0.867 (0.517–1.349)
XGBoost	0.747 (0.656–0.829)	0.685 (0.553–0.805)	0.199 (0.165–0.236)	-0.049 (-0.470–0.356)	0.859 (0.509–1.307)
RBF-SVM	0.751 (0.660–0.832)	0.672 (0.539–0.792)	0.199 (0.164–0.238)	0.020 (-0.419–0.430)	0.822 (0.484–1.252)
LightGBM	0.747 (0.655–0.830)	0.665 (0.527–0.796)	0.201 (0.177–0.227)	-0.050 (-0.422–0.290)	1.425 (0.856–2.134)

Higher AUROC and AP and lower Brier scores indicate better predictive performance. Ideal values are 0 for CITL and 1 for the calibration slope. Values are point estimates with 95% CIs. Internal validation includes 133 patients; external evaluation includes 102 hospitalization episodes from 50 patients. External confidence intervals accounted for repeated episodes within patients. AUROC, area under the receiver operating characteristic curve; AP, average precision; CITL, calibration-in-the-large. CI, confidence interval; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine.

Table 2. Predictive performance of the seven candidate models in internal and external evaluation.

B. External evaluation (102 admission-oriented episodes from 50 patients; 67 events)

Model	AUROC (95% CI)	Average precision (95% CI)	Brier score (95% CI)	CITL (95% CI)	Calibration slope (95% CI)
Logistic regression	0.712 (0.571–0.823)	0.785 (0.602–0.915)	0.225 (0.164–0.300)	0.781 (0.085–1.458)	0.223 (-0.038–0.774)
LASSO	0.705 (0.566–0.817)	0.776 (0.594–0.908)	0.225 (0.170–0.294)	0.717 (0.079–1.299)	0.255 (-0.064–0.962)
Elastic net	0.717 (0.577–0.828)	0.788 (0.607–0.916)	0.221 (0.170–0.285)	0.702 (0.101–1.239)	0.367 (-0.059–1.222)
Random forest	0.719 (0.567–0.838)	0.834 (0.639–0.934)	0.215 (0.160–0.282)	0.682 (0.101–1.266)	0.614 (0.198–1.142)
XGBoost	0.728 (0.569–0.846)	0.849 (0.646–0.941)	0.215 (0.162–0.278)	0.650 (0.085–1.215)	0.655 (0.208–1.210)
RBF-SVM	0.770 (0.640–0.871)	0.853 (0.707–0.945)	0.199 (0.147–0.259)	0.725 (0.159–1.272)	0.853 (0.415–1.491)
LightGBM	0.694 (0.537–0.820)	0.799 (0.592–0.920)	0.225 (0.191–0.267)	0.681 (0.146–1.169)	0.966 (0.227–1.825)

Higher AUROC and AP and lower Brier scores indicate better predictive performance. Ideal values are 0 for CITL and 1 for the calibration slope. Values are point estimates with 95% CIs. Internal validation includes 133 patients; external evaluation includes 102 hospitalization episodes from 50 patients. External confidence intervals accounted for repeated episodes within patients. AUROC, area under the receiver operating characteristic curve; AP, average precision; CITL, calibration-in-the-large. CI, confidence interval; LASSO, least absolute shrinkage and selection operator; RBF-SVM, radial-basis-function support vector machine; XGBoost, Extreme Gradient Boosting; LightGBM, Light Gradient Boosting Machine.